

SOC Home Lab

Detection Engineering & Incident Analysis Report

Simulated multi-stage attack — detection, correlation, and timeline reconstruction using Splunk

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Tooling: Splunk Enterprise (SIEM), Kali Linux, DVWA, Windows Sysmon, Hydra, Nmap, Atomic Red Team

Confidential — for portfolio / resume use

Executive Summary

This report documents a self-hosted Security Operations Center (SOC) home lab built to practice end-to-end detection engineering: generating realistic attack traffic, ingesting logs into Splunk, writing detection queries, and reconstructing an attacker's actions into a single correlated timeline. A Kali Linux attacker host was used to run a multi-stage attack chain against a target environment (a Linux/DVWA web server and a Windows endpoint instrumented with Sysmon). Splunk Enterprise ingested firewall (iptables), authentication, web server, and Windows Security/Sysmon event logs, which were then queried and correlated to reconstruct the attack from initial reconnaissance through to persistence.

Across a roughly one-hour window on August 4, 2026, the lab simulated a multi-stage attack sequence: reconnaissance (Nmap scanning), credential access (SSH brute force), web application exploitation (SQL injection against DVWA), endpoint execution (an encoded PowerShell command generated by Atomic Red Team), and finally privilege escalation and persistence on the Windows endpoint (local account creation, addition to Administrators, and a scheduled task). Seven detections were built as individual Splunk searches, each mapped to a MITRE ATT&CK technique, and stitched together below into a single chronological narrative.

Lab Environment

- SIEM / log aggregation: Splunk Enterprise (Search & Reporting)
- Attacker host: Kali Linux (Nmap, Hydra)
- Target — web application: Linux host running Apache2 + DVWA (Damn Vulnerable Web Application)
- Target — endpoint: Windows host instrumented with Sysmon, forwarding Security and Sysmon event logs
- Execution framework: Atomic Red Team (T1059.001 PowerShell test)
- Log sources ingested: /var/log/kern.log (iptables), /var/log/auth.log, /var/log/apache2/access.log, WinEventLog:Security, WinEventLog:Microsoft-Windows-Sysmon/Operational

Attack Chain Overview

The table below maps each stage of the attack to its MITRE ATT&CK technique and the Splunk log source used for detection.

#	Attack	MITRE	Log Source / Event	Key Evidence	Kill Chain
1	Nmap Port Scan	T1046	/var/log/kern.log — IPTABLES_SCAN	Source/dest IP, ports, protocol, time	Reconnaissance
2	SSH Brute Force	T1110.001	/var/log/auth.log	Source IP, username, failed→success sequence	Credential Access
3	SQL Injection	T1190	/var/log/apache2/access.log	Source IP, URI/payload, HTTP status, time	Initial Access
4	PowerShell Execution	T1059.001	Sysmon Event ID 1	User, process, parent, command line, GUID	Execution
5	Local User Creation	T1136.001	Security Event 4720	Creator, new account,	Persistence

#	Attack	MITRE	Log Source / Event	Key Evidence	Kill Chain
				SID, host, time	
6	Add User to Administrators	T1098	Security Event 4732	Subject, member, group, host, time	Privilege Escalation
7	Scheduled Task Creation	T1053.005	Security Event 4698	Creator, task name, task content, time	Persistence

Correlated Attack Timeline

The following timeline sequences all seven detections by event timestamp, showing how the attacker's actions moved from external reconnaissance to full Windows host compromise in under one hour.

Time (IST, UTC+05:30)	Tactic	Event	Technique	Severity
12:41:00	Reconnaissance	Nmap port scan	T1046	High
12:44:29 – 12:45:38	Credential Access	SSH brute force (Hydra) — 7 failed, 1 success	T1110.001	Critical
12:47:18 – 12:48:10	Initial Access	SQL injection against DVWA (incl. blind SQLi)	T1190	Critical
12:52:18 – 12:52:22	Execution	PowerShell — encoded command via cmd.exe (Atomic Red Team)	T1059.001	High
13:09:48	Persistence	Local account "attacker" created	T1136.001	High
13:18:08	Persistence	Local account "attacker" re-created	T1136.001	High
13:18:15	Privilege Escalation	"attacker" added to local Administrators group	T1098	Critical
13:40:14	Persistence	Scheduled task \SOC_Task created (trigger 23:59)	T1053.005	High

The timeline reads as a coherent multi-stage narrative rather than isolated alerts. Kali source 10.148.52.135 performed the Nmap reconnaissance, the SSH brute force against account "sujith," and the SQL injection activity against DVWA. Separately, the Windows endpoint recorded Atomic Red Team PowerShell execution followed by local account creation, an Administrator-group membership change, and scheduled-task creation. The telemetry supports these stages and their timestamps, but it does not independently prove that 10.148.52.135 performed the Windows-side actions directly — that link is inferred from timing and lab context, not from a captured network hop between the two hosts.

Detailed Findings by Attack Stage

1. Nmap Port Scan

MITRE ATT&CK: T1046 — Network Service Discovery

Tactic: Reconnaissance

Log Source: /var/log/kern.log (IPTABLES_SCAN)

Firewall logging on the target host recorded a burst of connection attempts across a wide range of destination ports from a single source, consistent with an Nmap scan used to enumerate open services before further exploitation.

SPL query used:

```
index=main source="/var/log/kern.log" IPTABLES_SCAN
| rex "SRC=(?<SourceIP>\d+\.\d+\.\d+\.\d+)"
| rex "DST=(?<DestinationIP>\d+\.\d+\.\d+\.\d+)"
| rex "DPT=(?<DestinationPort>\d+)"
| rex "PROTO=(?<Protocol>\w+)"
| bin _time span=1m
| stats dc(DestinationPort) as unique_ports values(DestinationPort) as ports_hit
  earliest(_time) as first_seen latest(_time) as last_seen by SourceIP
  DestinationIP _time
| eval first_seen=strftime(first_seen, "%Y-%m-%d %H:%M:%S")
| eval last_seen=strftime(last_seen, "%Y-%m-%d %H:%M:%S")
| where unique_ports > 15
```

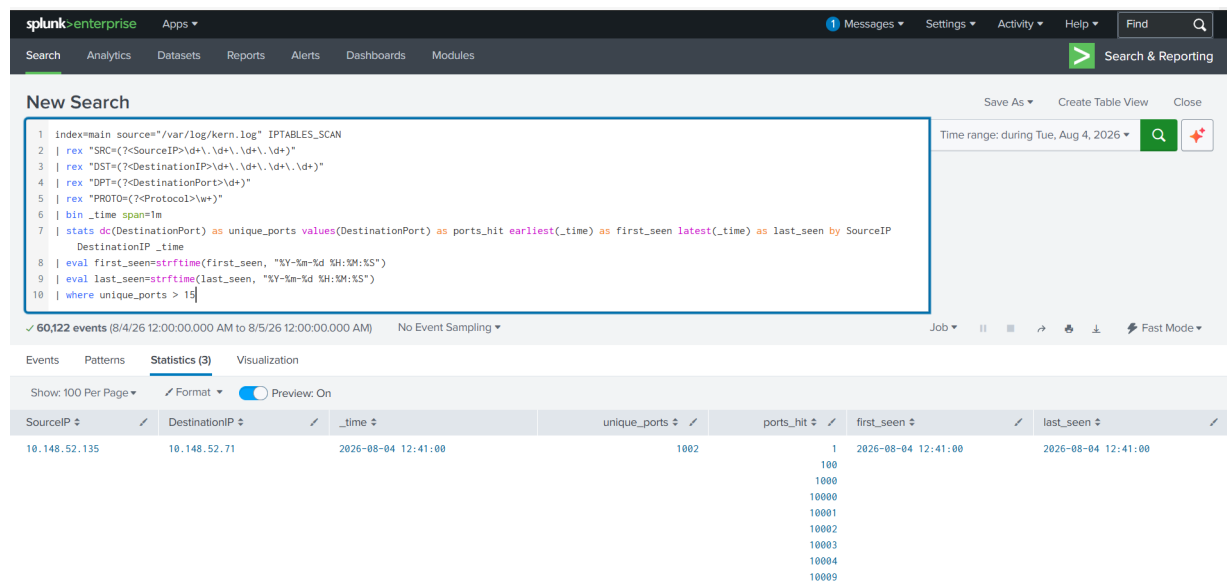


Figure 1: Splunk search results — Nmap port scan

Findings:

- Host 10.148.52.135 (attacker) sent traffic to 10.148.52.71 (target) touching 1,002 unique destination ports within a single one-minute bin — a strong signature of automated port scanning.
- iptables logging on the target captured the scan via kernel firewall log entries, giving a low-cost, host-based detection point that doesn't rely on network taps.
- First/last seen both resolve to 12:41:00, indicating the scan (or at least this burst) completed within the one-minute aggregation window.

2. SSH Brute Force

MITRE ATT&CK: T1110.001 — Password Guessing

Tactic: Credential Access

Log Source: /var/log/auth.log

A Hydra-driven brute-force attack against SSH was run from the same attacker host identified in the port scan. The query surfaces both the failed guesses and the eventual successful authentication, showing exactly when the attacker obtained valid credentials.

SPL query used:

```
index=main source="/var/log/auth.log" ("Failed password" OR "Accepted password")
| rex "(?<AuthenticationResult>Failed|Accepted) password for (?:(invalid user )?(?<Username>\S+))"
| from (?<SourceIP>\d+\.\d+\.\d+\.\d+) port (?<SourcePort>\d+)"
| eval Result=if(AuthenticationResult="Accepted","SUCCESS","FAILED")
| table _time SourceIP SourcePort Username Result host
| sort _time
```

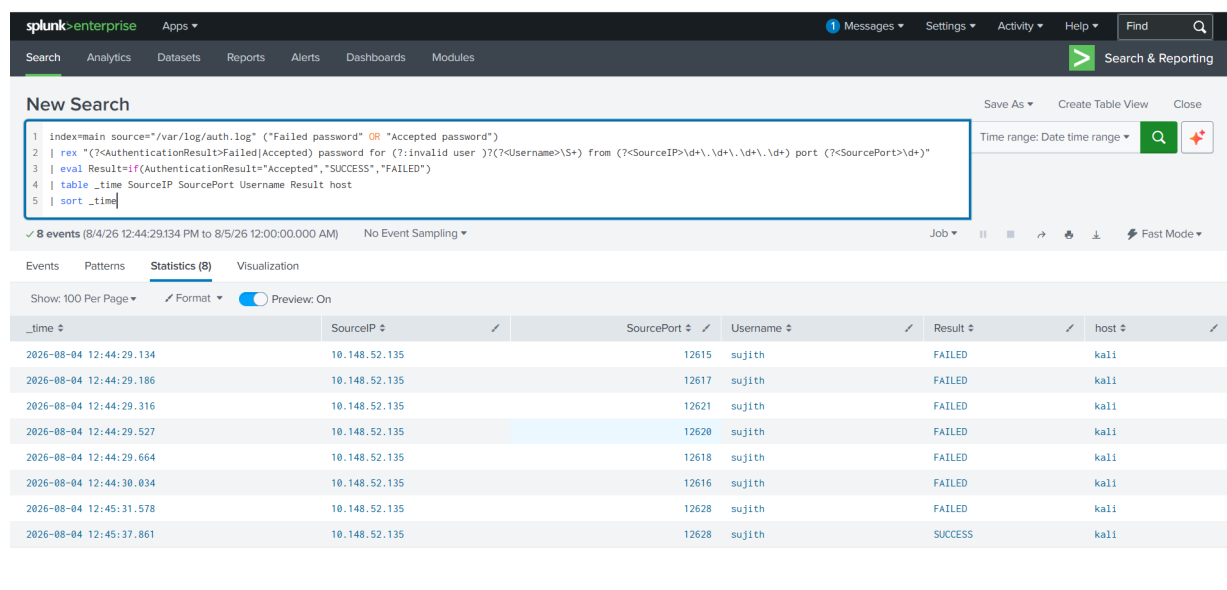


Figure 2: Splunk search results — SSH brute force

Findings:

- Eight authentication attempts against user "sujith" from 10.148.52.135 were captured between 12:44:29 and 12:45:37 on host "kali."
- Seven consecutive FAILED attempts (from sequential source ports 12615–12628, consistent with Hydra's connection handling) were followed by one SUCCESS at 12:45:37.861 — a textbook automated brute-force pattern.
- Ordering failed and successful attempts chronologically in a single table (rather than only alerting on the successful login) preserves the full attack story for the analyst and supports later correlation with the account's subsequent activity.

3. SQL Injection

MITRE ATT&CK: T1190 — Exploit Public-Facing Application

Tactic: Initial Access

Log Source: /var/log/apache2/access.log

With the target's web service already identified during reconnaissance, the attacker exercised DVWA's SQL injection module — first with baseline parameterized requests, then with an OR-based authentication bypass payload and a blind SQLi request.

SPL query used:

```
index=main source="/var/log/apache2/access.log" "/dwa/vulnerabilities/sqli/"
| rex "^(<SourceIP>\S+) \S+ \S+ \[(?<ApacheTime>[^\]]+)\]"
  "\"(?<Method>\S+) (?<URI>\S+) (?<Protocol>[^\"]+)\\" (?<Status>\d+) (?<Bytes>\d+)"
| table _time SourceIP Method URI Status Bytes
| sort _time
```

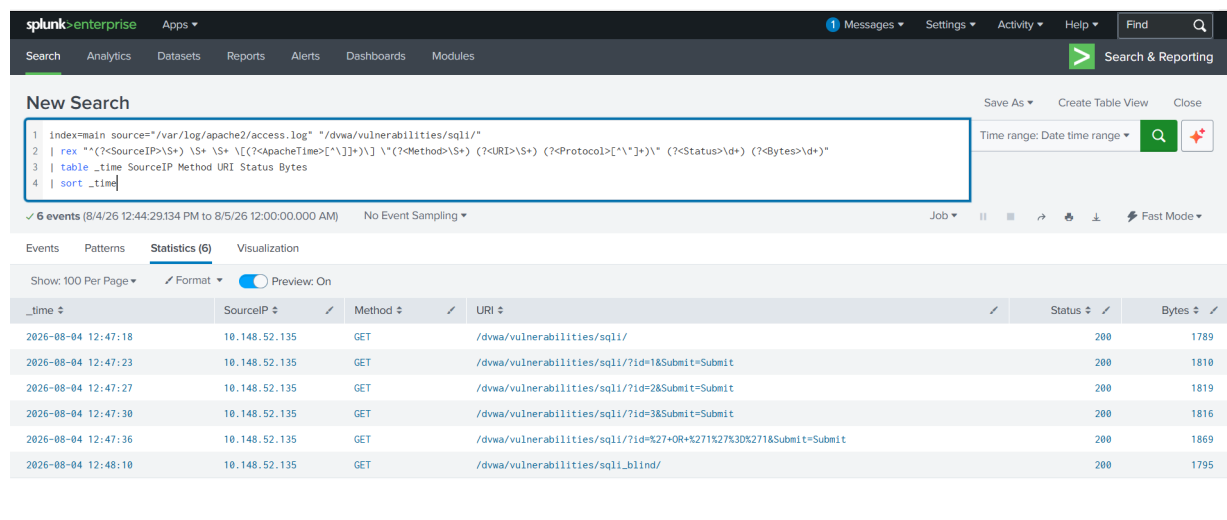


Figure 3: Splunk search results — SQL injection

Findings:

- Six requests to DVWA's SQL injection module were logged from 10.148.52.135 between 12:47:18 and 12:48:10, all returning HTTP 200.
- Requests escalated from baseline page loads (id=1, id=2, id=3) to a URL-encoded classic SQLi payload (id=%27+OR+%271%27%3D%271) at 12:47:36, followed by a request to the blind SQLi module at 12:48:10 — showing the attacker moving from manual testing into payload injection and blind exploitation.
- Correlating the requesting IP with the earlier Nmap/SSH activity confirms 10.148.52.135 as the same Kali source observed throughout. This establishes the SQL injection activity in the lab; it does not independently prove a subsequent pivot from DVWA to the Windows endpoint.

4. PowerShell Execution (Atomic Red Team)

MITRE ATT&CK: T1059.001 — PowerShell

Tactic: Execution

Log Source: Sysmon Event Code 1 (Process Creation)

Approximately four minutes after the web application was exploited, an encoded PowerShell command was executed on the Windows endpoint, launched from cmd.exe under the interactive user account — consistent with an attacker establishing code execution on the host.

SPL query used:

```
index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1
earliest="08/04/2026:12:52:00" latest="08/04/2026:12:53:00"
```

```

| rex "Image:\s+(?<ProcessImage>[^\r\n]+)"
| rex "ParentImage:\s+(?<ParentImage>[^\r\n]+)"
| rex "CommandLine:\s+(?<CommandLine>[^\r\n]+)"
| rex "User:\s+(?<User>[^\r\n]+)"
| search ProcessImage="*powershell.exe" OR Image="*powershell.exe"
| table _time User ProcessImage ProcessId ParentImage CommandLine ProcessGuid
| sort _time

```

New Search

```

1 index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1
2 earliest="08/04/2026:12:52:00" latest="08/04/2026:12:53:00"
3 | rex "Image:\s+(?<ProcessImage>[^\r\n]+)"
4 | rex "ProcessId:\s+(?<ProcessId>[^\r\n]+)"
5 | rex "ProcessGuid:\s+(?<ProcessGuid>[^\r\n]+)"
6 | rex "ParentImage:\s+(?<ParentImage>[^\r\n]+)"
7 | rex "CommandLine:\s+(?<CommandLine>[^\r\n]+)"
8 | rex "User:\s+(?<User>[^\r\n]+)"
9 | search ProcessImage="*powershell.exe" OR Image="*powershell.exe"
10 | table _time User ProcessImage ProcessId ParentImage CommandLine ProcessGuid
11 | sort _time

```

3 events (8/4/26 12:52:00.000 PM to 8/4/26 12:53:00.000 PM) No Event Sampling

Events Patterns **Statistics (3)** Visualization

Show: 100 Per Page Format Preview: On

_time	User	ProcessImage	ProcessId	ParentImage	CommandLine
2026-08-04 12:52:18.724	NT SERVICE\Splunkd	C:\Program Files\Splunk\bin\splunk-powershell.exe	11256	C:\Program Files\Splunk\bin\splunkd.exe	"C:\Program Files\Splunk\bin\splunk-powershell.exe"
2026-08-04 12:52:19.214	NT SERVICE\Splunkd	C:\Program Files\Splunk\bin\splunk-powershell.exe	13120	C:\Program Files\Splunk\bin\splunkd.exe	"C:\Program Files\Splunk\bin\splunk-powershell.exe" --ps2
2026-08-04 12:52:22.629	Sujith\SUJITH	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	23464	C:\Windows\System32\cmd.exe	powershell.exe -e JgAgACgAZwBjAGBIAAaCcaQBIASAMAB9ACCAIAATAGYIAAaHgaA3wpaCAkAIAA

Figure 4: Splunk search results — PowerShell execution (Atomic Red Team)

Findings:

- Scoping the search to the exact one-minute execution window (rather than filtering all EventCode=1 activity) isolated the relevant PowerShell process creation from routine Splunk service noise.
- At 12:52:22.629, user Sujith\SUJITH launched powershell.exe with an -e (EncodedCommand) argument and a Base64-encoded payload, spawned from cmd.exe under process ID 23464 — a pattern strongly associated with Atomic Red Team T1059.001 test cases and with real-world obfuscated PowerShell execution.
- The parent/child relationship (cmd.exe → powershell.exe -e <base64>) and full command line were preserved for the record, giving downstream investigators everything needed to decode and assess the payload.

5. Local User Creation

MITRE ATT&CK: T1136.001 — Local Account

Tactic: Persistence

Log Source: Windows Security Event Code 4720

Roughly 17 minutes after gaining code execution, the attacker used the compromised session to create a new local Windows account named "attacker" — a common persistence step that provides a durable foothold independent of the original entry vector.

SPL query used:

```

index=main source="WinEventLog:Security" EventCode=4720
| rex "Subject:\s*(?s).*?Account Name:\s+(?<SubjectUserName>[^\r\n]+)"
| rex "New Account:\s*(?s).*?Account Name:\s+(?<TargetUserName>[^\r\n]+)"
| rex "New Account:\s*(?s).*?Security ID:\s+(?<TargetSID>[^\r\n]+)"

```

```
| table _time SubjectUserName TargetUserName TargetSID host
| sort _time
```

The screenshot shows the Splunk search interface with a search bar containing the following query:

```
1 index=main source="WinEventLog:Security" EventCode=4720
2 | rex "Subject:\s*(?s).*?Account Name:\s+(?<SubjectUserName>[^\r\n]+)"
3 | rex "New Account:\s*(?s).*?Account Name:\s+(?<TargetUserName>[^\r\n]+)"
4 | rex "New Account:\s*(?s).*?Security ID:\s+(?<TargetSID>[^\r\n]+)"
5 | table _time SubjectUserName TargetUserName TargetSID host
6 | sort _time
```

The search results show 2 events from 8/4/26 12:00:00.000 AM to 8/5/26 12:00:00.000 AM. The table below displays the results:

_time	SubjectUserName	TargetUserName	TargetSID	host
2026-08-04 13:09:48.500	SUJITH	attacker	S-1-5-21-2050931781-1050032278-2459021037-1004	Sujith
2026-08-04 13:18:08.095	SUJITH	attacker	S-1-5-21-2050931781-1050032278-2459021037-1005	Sujith

Figure 5: Splunk search results — local user creation

Findings:

- Two 4720 events were recorded on host "Sujith": a local account named "attacker" was created at 13:09:48.500 (SID ending -1004), then the same account name was created again at 13:18:08.095 with a new SID (ending -1005). This shows the account name was created more than once; confirming whether the original account was deleted and recreated would need additional account-deletion telemetry.
- Both events show SUJITH as the subject (creating) account. This ties the account-creation action to the SUJITH security context; it does not independently prove that account was compromised beforehand.
- Capturing creator, new account name, SID, and host in one table gives a complete who-created-what-and-when record for the persistence mechanism.

6. Add User to Administrators Group

MITRE ATT&CK: T1098 — Account Manipulation

Tactic: Privilege Escalation

Log Source: Windows Security Event Code 4732

Within seconds of re-creating the local account, the attacker escalated it to full administrative rights by adding it to the built-in Administrators group, completing privilege escalation on the host.

SPL query used:

```
index=main source="WinEventLog:Security" EventCode=4732
| rex "Subject:\s*(?s).*?Account Name:\s+(?<SubjectUserName>[^\r\n]+)"
| rex "Member:\s*(?s).*?Account Name:\s+(?<MemberName>[^\r\n]+)"
| rex "Member:\s*(?s).*?Security ID:\s+(?<MemberSID>[^\r\n]+)"
| rex "Group:\s*(?s).*?Group Name:\s+(?<GroupName>[^\r\n]+)"
| search GroupName="Administrators"
| table _time SubjectUserName MemberName MemberSID GroupName host
| sort _time
```

New Search

```

1 index=main source='WinEventLog:Security' EventCode=4732
2 | rex 'Subject:\s*(?s).*?Account Name:\s+(?<SubjectUserName>[^\r\n]+)'
3 | rex 'Member:\s*(?s).*?Account Name:\s+(?<MemberName>[^\r\n]+)'
4 | rex 'Member:\s*(?s).*?Security ID:\s+(?<MemberSID>[^\r\n]+)'
5 | rex 'Group:\s*(?s).*?Group Name:\s+(?<GroupName>[^\r\n]+)'
6 | search GroupName='Administrators'
7 | table _time SubjectUserName MemberName MemberSID GroupName host
8 | sort _time

```

Time range: during Tue, Aug 4, 2026

1 event (8/4/26 12:00:00.000 AM to 8/5/26 12:00:00.000 AM) No Event Sampling

Events Patterns **Statistics (1)** Visualization

Show: 100 Per Page Format Preview: On

_time	SubjectUserName	MemberName	MemberSID	GroupName	host
2026-08-04 13:18:15.848	SUJITH	-	S-1-5-21-2050931781-1050032278-2459021037-1005	Administrators	Sujith

Figure 6: Splunk search results — add user to Administrators group

Findings:

- At 13:18:15.848 — seven seconds after the second account-creation event — SUJITH added the SID ending -1005 (the newly created "attacker" account) to the local Administrators group on host Sujith.
- This is the correct escalation event to alert on: Event 4732 (member added to a security-enabled local group) rather than 4672 (special privileges assigned to a logon), which only reflects privileges on an existing session.
- The tight timing between account creation (13:18:08) and group membership change (13:18:15) is itself a high-fidelity indicator worth alerting on as a combined detection rule.

7. Scheduled Task Creation

MITRE ATT&CK: T1053.005 — Scheduled Task

Tactic: Persistence

Log Source: Windows Security Event Code 4698

As a final persistence mechanism, the attacker registered a Windows scheduled task configured to fire at 23:59 the same day — ensuring continued access even if the interactive session or the newly created account were discovered and terminated in the interim.

SPL query used:

```

index=main source="WinEventLog:Security" EventCode=4698
| rex "Subject:\s*(?s).*?Account Name:\s+(?<SubjectUserName>[^\r\n]+)"
| rex "Task Name:\s+(?<TaskName>[^\r\n]+)"
| rex "Task Content:\s*(?<TaskContent>[^\s\S]*)"
| table _time SubjectUserName TaskName TaskContent host
| sort _time

```

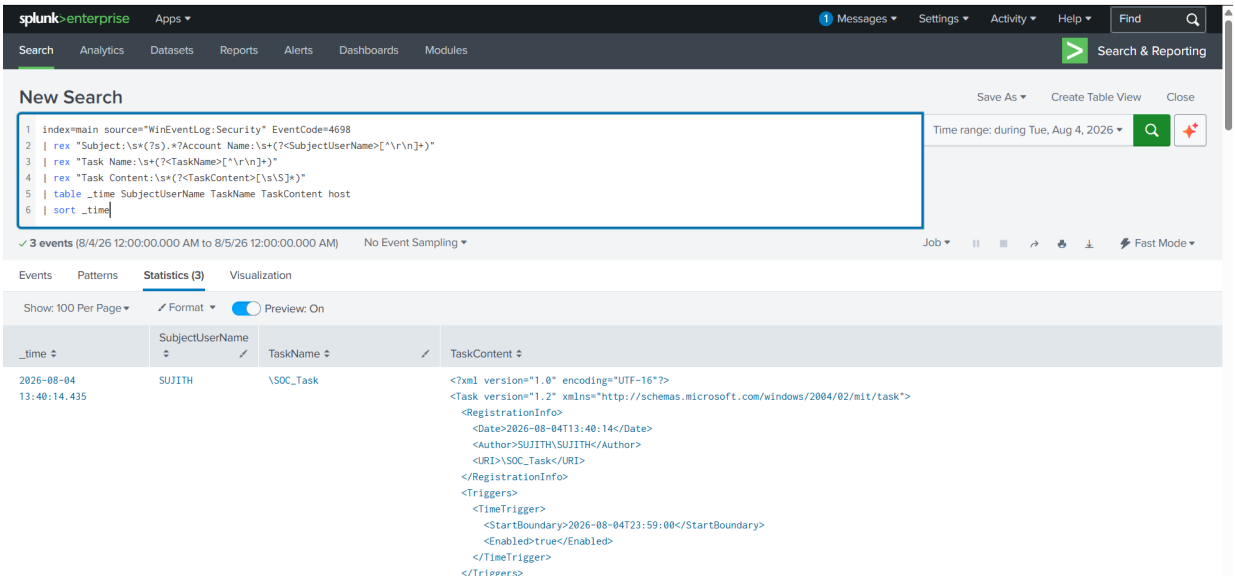


Figure 7: Splunk search results — scheduled task creation

Findings:

- At 13:40:14.435, SUJITH created a scheduled task named \SOC_Task on the Windows host "Sujith" — roughly 22 minutes after the Administrator-group membership change.
- The embedded Task Content XML shows a TimeTrigger with a StartBoundary of 2026-08-04T23:59:00, meaning the task is configured to execute later the same day. This is consistent with scheduled-task persistence or delayed execution and should be validated against the task action before being classified as malicious.
- Capturing the full task content in the table (creator, task name, and raw XML) gives responders everything needed to assess what the task actually runs without a separate pull from the endpoint.

Correlation Summary

All seven detections form a coherent multi-stage narrative across the Linux/DVWA and Windows environments. The first three stages are directly attributed to Kali source 10.148.52.135, while the Windows execution and post-exploitation stages are supported by local process and Windows Security telemetry. The available evidence does not independently establish a network-level pivot from the Kali IP to the Windows endpoint.

- 12:41 — Reconnaissance (Nmap) identifies the target and its exposed services.
- 12:44–12:45 — SSH brute force yields valid credentials for user "sujith."
- 12:47–12:48 — The same Kali source IP performs SQL injection activity against the web application.
- 12:52 — Atomic Red Team generates encoded PowerShell execution on the Windows endpoint.
- 13:09–13:18 — A new local account is created and escalated to Administrators.
- 13:40 — A scheduled task is registered for delayed, persistent execution.

Evidence integrity note: Kali source 10.148.52.135 is directly evidenced for the Nmap, SSH, and DVWA activity. The Windows stages are supported by local Sysmon and Windows Security events. This report does not claim an unsupported network pivot between the environments.

Skills Demonstrated

- Building detection content in Splunk SPL, including field extraction with rex, statistical aggregation (stats, bin), and result shaping (table, sort, eval).
- Mapping observed activity to the MITRE ATT&CK framework (T1046, T1110.001, T1190, T1059.001, T1136.001, T1098, T1053.005).
- Correlating multi-source logs (firewall, authentication, web server, Windows Security, Sysmon) into a single incident timeline.
- Simulating realistic attacker behavior end-to-end (Nmap, Hydra, DVWA, Atomic Red Team) to validate detections against known techniques.
- Writing analyst-ready findings that identify who did what, when, and how it was detected — the core structure of an incident report.

Recommendations

- Alert on the SSH brute-force pattern (5+ FAILED followed by SUCCESS from the same source within a short window) rather than on individual failed logins.
- Add a correlation rule linking Event 4720 (account creation) followed within minutes by Event 4732 (Administrators group addition) for the same host — this pairing is a strong privilege-escalation indicator.
- Alert on scheduled tasks (Event 4698) created by non-service accounts with a StartBoundary set for off-hours execution.
- Enable Sysmon Event ID 3 (network connections) in the endpoint configuration to support detection of C2 and exfiltration activity in future lab iterations.

Analyst Response Summary

The following summarizes the immediate SOC triage and response action for each confirmed detection.

Detection	Immediate Triage	Response / Next Step
Nmap	Validate source and scanned ports	Block/contain unauthorized source; review exposed services
SSH Brute Force	Confirm failed→success sequence and account	Reset credentials; block source; review SSH session
SQL Injection	Inspect URI/payload and HTTP response	Contain vulnerable app; review DB/app logs; remediate input handling
PowerShell	Review command line, parent, user and ProcessGuid	Validate Atomic test; isolate if unauthorized; hunt related processes
4720 User Creation	Identify creator and new SID	Validate authorization; disable/remove unauthorized account
4732 Administrators	Confirm member and group	Remove unauthorized membership; review privileged activity
4698 Scheduled Task	Inspect TaskName, TaskContent and trigger	Validate task; remove unauthorized task; hunt for persistence